## ■ 1.5.6 Advanced Topic: Logical Operations

| | |
|---|---|
| *x* `==` *y* | equal |
| *x* `!=` *y* | unequal |
| *x* `>` *y* | greater than |
| *x* `>=` *y* | greater than or equal |
| *x* `<` *y* | less than |
| *x* `<=` *y* | less than or equal |
| *x* `==` *y* `==` *z* | all equal |
| *x* `!=` *y* `!=` *z* | all unequal (distinct) |
| *x* `>` *y* `>` *z*, etc. | strictly decreasing, etc. |

Relational operators.

This tests whether `10` is less than `7`. The result is `False`.

```
In[1]:= 10 < 7
Out[1]= False
```

Not all the numbers are unequal, so this gives `False`.

```
In[2]:= 3 != 2 != 3
Out[2]= False
```

You can mix `<` and `<=`.

```
In[3]:= 3 < 5 <= 6
Out[3]= True
```

*Mathematica* does not know whether this is true or false.

```
In[4]:= x > y
Out[4]= x > y
```

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| | |
|---|---|
| `!`$p$ | not |
| $p$ `&&` $q$ `&&` ... | and |
| $p$ `\|\|` $q$ `\|\|` ... | or |
| `Xor[`$p$`,` $q$`,` ...`]` | exclusive or |
| `Implies[`$p$`,` $q$`]` | implication $p \Rightarrow q$ |
| `If[`$p$`,` $t$`,` $f$`]` | give $t$ if $p$ is `True`, and $f$ if $p$ is `False` |
| `LogicalExpand[`*expr*`]` | expand out logical expressions |

Logical operations.

Both tests give `True`, so the result is `True`.

```
In[5]:= 7 > 4 && 2 != 3
Out[5]= True
```

You should remember that the logical operations `==`, `&&` and `||` are all *double characters* in *Mathematica*. (If you have used the C programming language, you will recognize this notation as being the same as in C.)

*Mathematica* does not know whether this is true or false.

```
In[6]:= p && q
Out[6]= p && q
```

*Mathematica* leaves this expression unchanged.

```
In[7]:= (p || q) && !(r || s)
Out[7]= (p || q) && !(r || s)
```

You can use `LogicalExpand` to expand out the terms.

```
In[8]:= LogicalExpand[ % ]
Out[8]= p && !r && !s || q && !r && !s
```

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